

ICML

1 Safe Stabilization with Neural Control Density Functions

1.1 Preliminaries

We consider dynamical systems of the form

$$\dot{x} = f(x) + g(x)u(x) \quad (1)$$

We assume here that f and g are analytic.

Next, we re-state the result of Masubuchi, which guarantees local stability and forward invariance.

Theorem 1.1 (Masubuchi [2009], Theorem 1). *Let S be a compact set, and suppose the system*

$$\dot{x} = f(x)$$

has $f(x)$ analytic. If there exists a $\rho : \mathbb{R}^n \rightarrow (0, K]$ for some $K < \infty$ that satisfies

$$\nabla \cdot [f\rho] > 0$$

then for almost all initial conditions $x_0 \in S$, trajectories converge to the origin.

1.2 The Main Result

Theorem 1.2. *Let S be a bounded set such that $0 \in S$. Consider the dynamical system (1), and define the set $\mathcal{C} := \{x \in S : h(x) > 0\} \subset S$, where $h \in C^1(\mathbb{R}^n, \mathbb{R})$. Suppose there exists a positive, analytic density function ρ and $\psi \in C^\infty(\mathbb{R}^n, \mathbb{R})$ such that $f(x)\frac{\rho(x)}{\|x\|} + g(x)\frac{\psi(x)}{\|x\|}$ is integrable on $\{x \in \mathbb{R}^n : |x| \geq 1\}$, and suppose there exists an extended class $\mathcal{K}_\infty$ function α such that:*

$$\begin{aligned} \nabla \cdot (\rho(x)f(x) + \psi(x)g(x)) &> 0 \quad \text{for almost all } x \in S \\ \rho(x)(L_f h(x) + \alpha(h(x))) + \psi(x)L_g h(x) &\geq 0 \quad \text{for almost all } x \in S. \end{aligned} \quad (2)$$

for all $x \in \mathcal{C}$. Then, there exists a static state feedback controller $u(x)$ such a solution $x(t)$ of (1), starting from almost all $x_0 \in \mathcal{C}$, satisfies $\lim_{t \rightarrow \infty} x(t) = 0$ and, for all $x \in \mathcal{C}$, $\inf_{t \geq 0} x(t) \in \mathcal{C}$.

Proof Sketch. Since we assume f, g to be analytic as well, and since $\rho > 0$, we also get $F(x) = f(x) + g(x)\psi(x)/\rho(x)$ is analytic. This satisfies the conditions of Theorem 1.1. The rest of the proof follows as shown in Appendix C.4. $\square$

Remark 1.3. It must be noted that the neural control density functions obtained via Proposition 4.1 yields analytic ρ and ψ since analytic activation functions are used.

2 Explanation of (18) to (19)

We consider the following parameterization:

$$\rho(x) = \frac{a(x)}{\exp(\|x\|^2 + \|b(x)\|^2)} \quad \psi(x) = \frac{c(x)}{\exp(\|x\|^2 + \|b(x)\|^2)} \quad (3)$$

The divergence condition requires:

$$\begin{aligned}
\nabla \cdot [\rho f + g\psi] &= \nabla \cdot \left[\frac{af + gc}{\exp(\|x\|^2 + \|b(x)\|^2)} \right] \\
&= \frac{1}{\exp(\|x\|^2 + \|b(x)\|^2)} \nabla \cdot [af + gc] - (af + gc)^T (2x + \nabla \|b(x)\|^2) \frac{1}{\exp(\|x\|^2 + \|b(x)\|^2)} \\
&= \frac{\nabla \cdot [af + gc] - (af + gc)^T (2x + \nabla \|b(x)\|^2)}{\exp(\|x\|^2 + \|b(x)\|^2)}
\end{aligned}$$

Since, $\exp(\cdot) \in \mathbb{R}_{>0}$, it suffices to check if

$$\nabla \cdot [af + gc] - (af + gc)^T (2x + \nabla \|b(x)\|^2) > 0 \quad (4)$$

3

Ducted fan: The dynamics is given by:

$$\begin{aligned}
\ddot{x} &= \frac{1}{m}(-d\dot{x} + u_0 \cos \theta - u_1 \sin \theta) \\
\ddot{y} &= \frac{1}{m}(-d\dot{y} + u_0 \sin \theta + u_1 \cos \theta) - g \\
\ddot{\theta} &= \frac{r}{I}u_0
\end{aligned}$$

with $m = 11.2, r = 0.156, I = 0.0462, d = 0.1$ and $g = 0.28$ where the state variables is $[x \ y \ \theta \ \dot{x} \ \dot{y} \ \dot{\theta}]$.

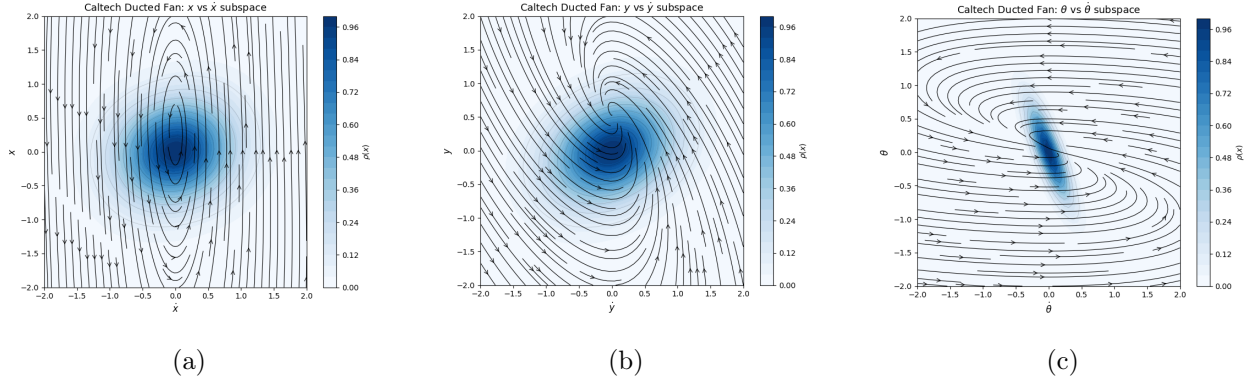


Figure 1: (a)–(c): Contour for various subspace of caltech ducted fan

PVTOL: The dynamics is given by:

$$\begin{bmatrix} p_x \\ p_z \\ v_x \\ v_z \\ \phi \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} v_x \cos \phi - v_z \sin \phi \\ v_x \sin \phi + v_z \cos \phi \\ \dot{\phi} \\ v_z \dot{\phi} - g \sin \phi \\ -v_x \dot{\phi} - g \cos \phi \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & \frac{1}{m} \\ \frac{1}{J} & \frac{1}{J} \end{bmatrix} u \quad (5)$$

where $g = 9.8, m = 4, l = 0.25, J = 0.0475$

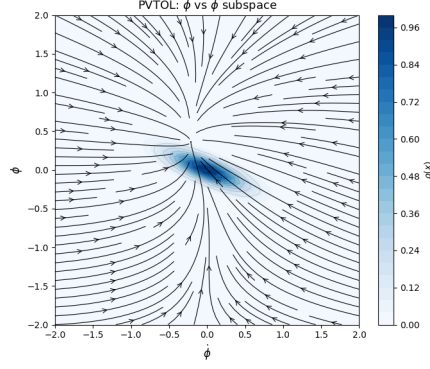


Figure 2: Contour of PVTOL system

4 Computational Study

Learning and Verification time: We provide the time to learn control-density pair for various system in the table below. We fix the falsifier call to every 100 iterations with learning rate $1e-3$.

S.N	Dynamics	Number of parameters	Time
1	Inverted Pendulum	A-net: 109, B-net: 96, C-net:109	3m 23s
2	Path Following	A-net: 13, B-net: 12, C-net:49	3m 11s
3	Caltech Ducted Fan	A-net: 2369,B-net: 2534, C-net:2402	21m 24s

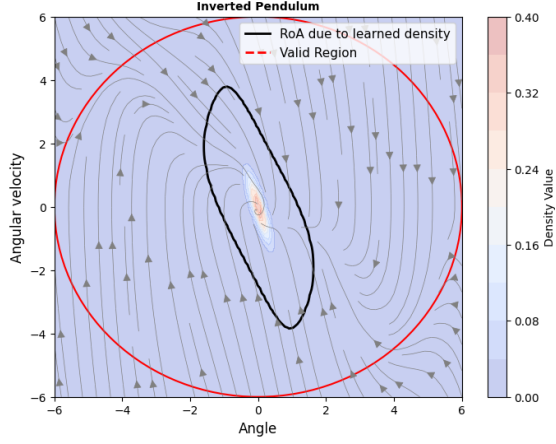
Table 1: Time taken to learn the control-density pair

Total time vs Falsification Frequency: We use an inverted pendulum model to investigate how the total learning time for a valid control-density pair varies with the falsification frequency.

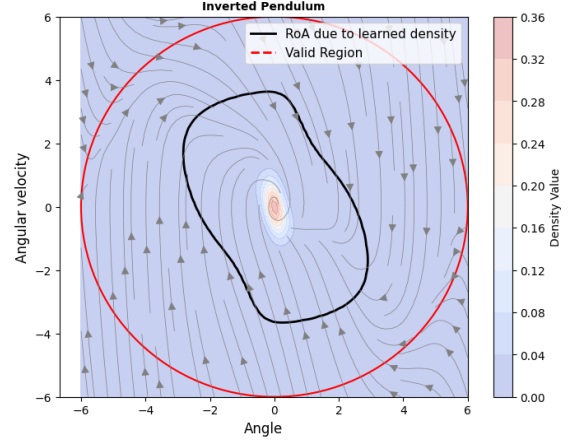
S.N	Frequency of Falsifier	Time
1	100	3m 23s
2	250	3m 53s
3	500	23s
4	750	22.4s

Table 2: Time taken to learn the control-density pair varying the frequency the falsifier calls

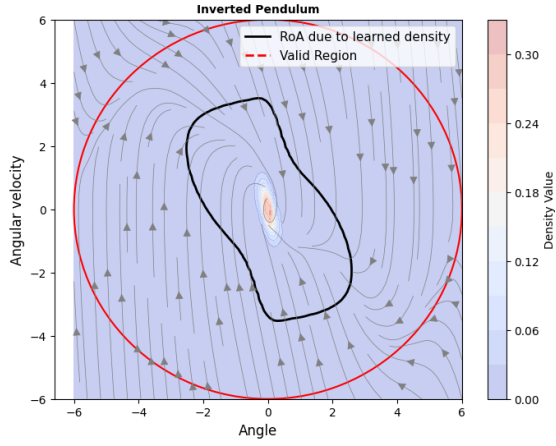
The contour corresponding to inverted pendulum with various falsification frequency is provided below:



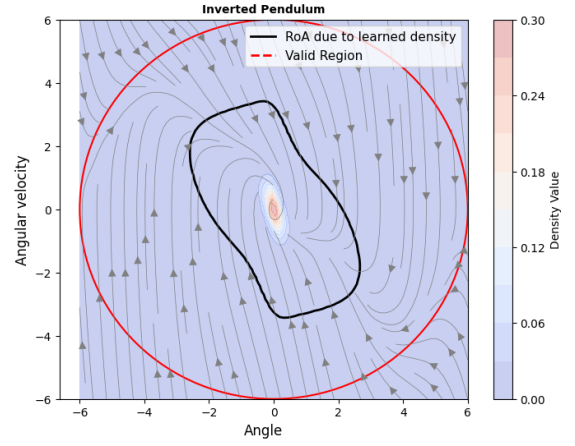
(a): Falsifier called every 100 iterations



(b): Falsifier called every 250 iterations



(c): Falsifier called every 500 iterations



(d): Falsifier called every 750 iterations

Figure 3: Contour for Inverted Pendulum

References

Izumi Masubuchi. Stability analysis of nonlinear systems via locally defined density functions. In *2009 ICCAS-SICE*, pages 2454–2457, 2009.